ℓ^1-Native Cohomology of Finite Defect Fields

Quotient Norms, Dual Circulations, Sparse Representatives, and the L^p Geometric Family

Jeremy H. Carroll

2026-10-15

License: Creative Commons Attribution 4.0 International (CC-BY-4.0)

Scope: Finite-dimensional convex and combinatorial graph cohomology.

Version: 6.5.2 (2026-10-15)

Notation and Mathematical Setup

We fix standard notation for finite cell complexes and convex analysis used throughout this monograph:

- $G = (V, E)$: A finite directed graph with $n = |V|$ vertices and $m = |E|$ edges. Edge orientations are fixed for bookkeeping.
- $C^0 = \mathbb{R}^V$, $C^1 = \mathbb{R}^E$: Finite-dimensional vector spaces of 0-cochains (vertex potentials) and 1-cochains (edge differences).
- $d_0 : C^0 \to C^1$: Incidence coboundary operator, $(d_0 f)(u, v) = f(v) - f(u)$ for directed edge $e = (u, v) \in E$, represented by the $m \times n$ incidence matrix $\mathbf{B}$.
- $B^1 = \operatorname{im} d_0$: The subspace of exact 1-cochains (pure gradient potentials).
- $Q^1 = C^1/B^1$: The first cochain quotient space of dimension $\beta = m - n + c$, where c is the number of connected components (the cycle rank or first Betti number of G).
- $\mathbf{w} \in \mathbb{R}^E_{>0}$: A strictly positive edge weight vector.
- $\Phi_p([\omega]) = \inf_{f \in C^0} \|\omega + d_0 f\|_{p,\mathbf{w}}$: The weighted quotient L^p norm for $p \in [1, \infty]$.
- $Z_1(G; \mathbb{R}) = \ker(\mathbf{B}^\top)$: Real divergence-free circulations, used for the LP dual. Integral cycles $Z_1(G; \mathbb{Z}) = \ker(\mathbf{B}^\top) \cap \mathbb{Z}^E$ are used for branch-independent multiplicative holonomy.
- For 2-simplicial complexes $K = (V, E, F)$, $d_1 : C^1 \to C^2$ denotes the face coboundary operator, satisfying discrete Bianchi nilpotency $d_1 \circ d_0 = 0$.

Abstract

For a finite directed graph with strictly positive edge weights, we study the family of weighted L^p norms ($p \in [1, \infty)$, complemented by the separate weighted Chebyshev endpoint norm at $p = \infty$) induced on the first cochain quotient space $Q^1 = C^1/\operatorname{im} d_0$. The quotient infimum is attained for all $p \in [1, \infty]$, defines a genuine quotient norm Φ_p, and satisfies Fenchel-Rockafellar duality with conjugate exponent $q = p/(p-1)$, characterizing the quotient norm as a maximum circulation pairing subject to dual L^q capacity bounds. We establish the geometric transition across regimes: (i) $p = 1$ is polyhedral and non-strictly convex, may admit non-unique primal minimizers, exhibits sparse cycle-rank support bounds ($|\operatorname{supp}(r)| \le \beta$), and admits exact rational linear programming duality; (ii) for $p \in (1, \infty)$ the weighted cochain normed space is uniformly convex, and the powered objective $J_p(f) = \frac{1}{p} \sum_e w_e |\omega_e + (\mathbf{B}f)_e|^p$ is strictly convex on gauge-fixed potentials (one zero potential per connected component), ensuring a strictly unique optimal residual $\mathbf{r}_p^* = \omega + \mathbf{B}\mathbf{f}_p^*$ and unique gauge-fixed potential $\mathbf{f}_p^*$ characterized by nonlinear graph p-Laplacian optimality $\mathbf{B}^\top(\mathbf{w} \odot |\mathbf{r}|^{p-1} \odot \operatorname{sgn} \mathbf{r}) = \mathbf{0}, \mathbf{r} = \omega + \mathbf{B}\mathbf{f}$; (iii) $p = 2$ recovers classical Hilbertian Hodge-Helmholtz orthogonal decomposition with Moore-Penrose graph Laplacian solutions and weighted dual circulation witnesses $\mathbf{z}_2^* = \mathbf{W}\mathbf{r}_2^*/\|\mathbf{r}_2^*\|_{2,\mathbf{w}}$, where $\mathbf{W} = \operatorname{diag}(\mathbf{w})$, when $\mathbf{r}_2^* \neq \mathbf{0}$ (with $\mathbf{z}_2^* = \mathbf{0}$ at zero residual); (iv) $p = \infty$ corresponds to discrete

Chebyshev approximation governed by active-edge subgradient pairing under its separate weighted endpoint convention $\|x\|_{\infty,\mathbf{w}} = \max_e w_e|x_e|$ (with the general Aronsson equation withdrawn for arbitrary cochains); and (v) the $p \to 1^+$ limit is characterized via unconditional secondary entropy minimization over the optimal L^1 face. We also show that simplicial coboundary operators satisfy discrete Bianchi nilpotency with gauge-orbit circulation invariance, that off-diagonal quantum state coherence is bounded under Schatten trace norms, and that additive real flux $\Phi_1([\omega])$ of a chosen real lift is distinguished from branch-independent $U(1)$ phase holonomy and period-minimized quotient defect. These general finite-dimensional statements have prose proofs in the manuscript and are classified as classical background. Lean mechanizes the conditional finite integer certificate L1C-TG and the specified finite-model lemmas for L1C-T1, L1C-T2, L1C-T3, L1C-T4, L1C-T5, and L1C-T7; this formalization scope is distinct from mathematical proof and originality.

Jiang, Lim, Yao, and Ye [9, preprint v2, §6.1] already formulate weighted L^1 projection of edge flows onto gradients and its dual over bounded divergence-free flows. Here the contribution is a presentation through weighted finite-graph quotient optimization and exact, independently checkable certificates, a rational computational companion, and mechanization of specified finite integer certificate statements in Lean. The bounded-circulation dual is classical background.

Why a researcher should care

Least squares ($p = 2$) spreads error globally over all edges. L^1 quotient minimization instead localizes defect on sparse cycle representatives while retaining an exact rational dual certificate. The complete L^p family connects sparse polyhedral localization ($p = 1$), strictly unique nonlinear p-Laplacian projections ($p \in (1, \infty)$), classical Hilbertian Hodge theory ($p = 2$), and maximal-error Chebyshev suppression ($p = \infty$, under its separate weighted endpoint convention). The certificate presentation connects these classical graph and convex constructions with machine-checkable finite witnesses.

Falsifiability

The prose propositions and the separately scoped Lean theorems have the hypotheses stated in their sections and claim records. A mathematical conclusion would fail if a positive-weight finite instance has no attained minimum for some $p \in [1, \infty]$, if primal and dual values differ for an optimal pair, if data satisfying all stated hypotheses of Theorem L1C-T4 or LpC-T3 fail their conclusions, if for $p \in (1, \infty)$ two distinct gauge-fixed potentials (one fixed vertex per component) both minimize the quotient objective, if the nonlinear p-Laplacian Euler-Lagrange equation fails at an optimal potential, if some class lacks an L^1 optimum with support at most the cycle rank, if an integral instance has no integral L^1 optimum, if the $p \to 1^+$ trajectory fails to converge to an L^1 minimizer, if a 2-simplicial complex has non-vanishing field curvature $d_1(d_0 f) \neq 0$ for a pure gauge potential, or if a finite quantum state violates the off-diagonal Schatten-p or trace coherence bound. The companion code and tests check solver success, primal/dual conditions, support bounds, exact-shift invariance, nilpotency, and quantum coherence bounds across enumerated instances; it also includes an instance on which an L^2 surrogate reports the wrong L^1 value.

1. Finite graph cochains and L^p norms

Let $G = (V, E)$ be a finite directed graph. Orientation is bookkeeping: reversing an edge negates its cochain coordinate. Put

$$C^0 = \mathbb{R}^V, \quad C^1 = \mathbb{R}^E, \quad \text{and} \quad (d_0 f)(u, v) = f(v) - f(u).$$

When considering 1-dimensional graph carriers without higher faces, $d_1 = 0$, so every edge cochain is closed, while exact cochains form $B^1 = \operatorname{im} d_0$. The first cochain quotient is

$$Q^1 = C^1 / B^1.$$

Fix weights $w_e > 0$ for all $e \in E$. For $p \in [1, \infty)$, the weighted L^p norm on C^1 is

$$\|x\|_{p,\mathbf{w}} = \left(\sum_{e \in E} w_e |x_e|^p \right)^{1/p},$$

and for $p = \infty$, the weighted L^∞ norm is defined under the Chebyshev convention

$$\|x\|_{\infty,\mathbf{w}} = \max_{e \in E} w_e |x_e|.$$

Remark on the $p = \infty$ endpoint weighting convention. For non-uniform weights $\mathbf{w}$, the Chebyshev endpoint norm $\|x\|_{\infty,\mathbf{w}} = \max_{e \in E} w_e |x_e|$ is a separate weighted endpoint convention chosen for maximal weighted coordinate error suppression (standard in discrete Chebyshev approximation and computational engineering). It does not coincide with the point-wise analytic limit $\lim_{p \to \infty} \|x\|_{p,\mathbf{w}} = \max_{e \in E} |x_e|$ of the finite-p weighted norms $(\sum w_e |x_e|^p)^{1/p}$. On an unweighted graph ($w_e = 1$ for all e), the two conventions coincide and the finite-p norms interpolate continuously to ℓ^∞. For non-uniform weights (for example, on a triangle with cycle flux 1 and weights $(1, 2, 3)$, where $\lim_{p \to \infty} \Phi_p = 1/3$ whereas the defined Chebyshev defect is $\Phi_\infty = 6/11$), the $p = \infty$ problem is treated as a separate polyhedral endpoint convention with its own active-edge subgradient pairing (see §4.4).

For $p = 1$, this specializes to the polyhedral weighted L^1 norm $\|x\|_{1,\mathbf{w}} = \sum_{e \in E} w_e |x_e|$. For a quotient class $[\omega] \in Q^1$ and $p \in [1, \infty]$, set

$$\Phi_p([\omega]) = \inf_{f \in C^0} \|\omega + d_0 f\|_{p,\mathbf{w}}.$$

All results extend componentwise to disconnected graphs. For concise proofs we assume G is connected and write $n = |V|$, $m = |E|$, and $\beta = m - n + 1 = \dim Q^1$.

2. Attainment and quotient norm

Theorem LpC-T1 — Lp quotient attainment

For every $p \in [1, \infty]$ and every $\omega \in C^1$, the infimum defining $\Phi_p([\omega])$ is attained.

Proof. Fix a root vertex $v_0 \in V$ and restrict to the gauge-fixed subspace $F = \{f \in C^0 : f(v_j) = 0 \text{ for one root } v_j \text{ per component}\}$. Connectedness makes $d_0|_F$ injective. In finite dimension, all norms are equivalent and an injective linear map is bounded below: for any $p \in [1, \infty]$, there exists $c_p > 0$ such that $\|d_0 f\|_{p,\mathbf{w}} \geq c_p \|f\|$ for all $f \in F$. Hence

$$\|\omega + d_0 f\|_{p,\mathbf{w}} \geq c_p \|f\| - \|\omega\|_{p,\mathbf{w}},$$

so the objective is coercive on F. Because $f \mapsto \|\omega + d_0 f\|_{p,\mathbf{w}}$ is continuous on the finite-dimensional space F, it attains its minimum. Attainment on F implies attainment on C^0 because $C^0 = F \oplus \ker(d_0)$.

□

Theorem L1C-T1 — attainment (p = 1 base case)

For every $\omega \in C^1$, the infimum defining $\Phi_1([\omega])$ is attained.

Proof. Specializing the classical argument of Theorem LpC-T1 to $p = 1$ yields coercivity of $f \mapsto \|\omega + d_0 f\|_{1,\mathbf{w}}$ on F. Lean 4 elaborates the integer 3-cycle bound in `lean4/l1c_t01_attainment.lean`.

□

Theorem LpC-T2 — Lp quotient norm

For every $p \in [1, \infty]$, Φ_p is a norm on $Q^1 = C^1/\text{im } d_0$.

Proof. Independence of representative follows by translating $f \mapsto f + g$. Homogeneity holds because for $\lambda \in \mathbb{R}$, $\Phi_p([\lambda\omega]) = \inf_f \|\lambda\omega + d_0 f\|_{p,\mathbf{w}} = |\lambda| \inf_{f'} \|\omega + d_0 f'\|_{p,\mathbf{w}} = |\lambda|\Phi_p([\omega])$. Subadditivity follows from Minkowski's inequality: for $[\omega_1], [\omega_2] \in Q^1$ with respective minimizers f_1, f_2,

$$\Phi_p([\omega_1 + \omega_2]) \leq \|\omega_1 + \omega_2 + d_0(f_1 + f_2)\|_{p,\mathbf{w}} \leq \|\omega_1 + d_0 f_1\|_{p,\mathbf{w}} + \|\omega_2 + d_0 f_2\|_{p,\mathbf{w}} = \Phi_p([\omega_1]) + \Phi_p([\omega_2]).$$

Positive definiteness: If $\Phi_p([\omega]) = 0$, attainment (Theorem LpC-T1) provides f^* such that $\|\omega + d_0 f^*\|_{p,\mathbf{w}} = 0$. Because each $w_e > 0$, this implies $\omega + d_0 f^* = 0$, so $\omega \in \text{im } d_0$, meaning $[\omega] = 0$. Conversely, if $\omega \in \text{im } d_0$, taking $f = -d_0^\dagger \omega$ gives norm zero.

□

Theorem L1C-T2 — quotient norm (p = 1 base case)

Φ_1 is a norm on Q^1.

Proof. Specialize Theorem LpC-T2 to $p = 1$. Lean 4 separately elaborates the scoped ambient weighted-norm algebra in `lean4/l1c_t02_quotient_norm.lean`. Strict positivity of every w_e is essential: allowing a zero weight permits a nonzero quotient class to register value zero.

□

3. Fenchel-Rockafellar Hölder duality

Let $\mathbf{B}$ denote the $m \times n$ incidence matrix representing d_0. The primal problem for $p \in [1, \infty]$ is:

$$\inf_{\mathbf{f} \in \mathbb{R}^n} \|\omega + \mathbf{B}\mathbf{f}\|_{p,\mathbf{w}}.$$

Define the conjugate exponent $q \in [1, \infty]$ by $1/p + 1/q = 1$ (with $q = \infty$ for $p = 1$, and $q = 1$ for $p = \infty$). The dual norm on 1-chains $\mathbf{z} \in \mathbb{R}^E$ is defined by:

$$\|\mathbf{z}\|_{q,\mathbf{w}^*} = \left(\sum_{e \in E} w_e^{1-q} |z_e|^q \right)^{1/q} \quad \text{for } 1 < q < \infty,$$

$$\|\mathbf{z}\|_{\infty,\mathbf{w}^*} = \max_{e \in E} \frac{|z_e|}{w_e} \quad \text{for } q = \infty \ (p = 1),$$

$$\|\mathbf{z}\|_{1,\mathbf{w}^*} = \sum_{e \in E} \frac{|z_e|}{w_e} \quad \text{for } q = 1 \ (p = \infty).$$

Theorem LpC-T3 — Fenchel-Rockafellar Hölder duality

For every $p \in [1, \infty]$ and conjugate exponent $q = p/(p-1)$, strong duality holds:

$$\Phi_p([\omega]) = \max_{\mathbf{z} \in \mathbb{R}^E} \left\{ \mathbf{z}^\top \omega : \mathbf{B}^\top \mathbf{z} = 0, \quad \|\mathbf{z}\|_{q,\mathbf{w}^*} \leq 1 \right\}.$$

Both primal and dual extrema are attained.

Proof. Use the gauge-fixed subspace for primal attainment (Theorem LpC-T1). Let $F(\mathbf{u}) = \|\mathbf{u}\|_{p,\mathbf{w}}$ denote the weighted norm and $H(\mathbf{u}) = F(\omega + \mathbf{u})$. Its Fenchel conjugate is $H^*(\mathbf{z}) = F^*(\mathbf{z}) - \mathbf{z}^\top \omega$, where F^* is the indicator function of the dual unit ball $\{\mathbf{z} \in \mathbb{R}^E : \|\mathbf{z}\|_{q,\mathbf{w}^*} \leq 1\}$. Because H is finite and continuous on

$\mathbb{R}^E$, finite-dimensional Fenchel-Rockafellar duality [1] equates the primal minimum over $\mathbf{Bf}$ with the dual maximum of $\mathbf{z}^\top\omega$ over $\mathbf{B}^\top\mathbf{z} = \mathbf{0}$ and $\|\mathbf{z}\|_{q,\mathbf{w}^*} \leq 1$, using norm symmetry. Compactness of this dual feasible set yields attainment.

□

Theorem L1C-T3 — dual circulation formula (p = 1 base case)

For $p = 1$, the dual norm constraint $\|\mathbf{z}\|_{\infty,\mathbf{w}^*} \leq 1$ is equivalent to coordinate capacity bounds $|z_e| \leq w_e$, yielding:

$$\Phi_1([\omega]) = \max_{\mathbf{z}} \left\{ \mathbf{z}^\top\omega : \mathbf{B}^\top\mathbf{z} = 0, \quad |z_e| \leq w_e \quad \forall e \in E \right\}.$$

Both extrema are attained.

Proof. The specialization of Theorem LpC-T3 to $p = 1$ gives strong LP duality over bounded circulations. Lean 4 elaborates the 3-cycle weak-duality inequality in `lean4/l1c_t03_weak_duality.lean`.

□

Corollary LpC-C1 — Lp exactness criterion

For any $p \in [1, \infty]$, ω is exact if and only if every circulation pairs to zero with ω.

Proof. If $\omega = \mathbf{Bf}$, then for any $\mathbf{z}$ with $\mathbf{B}^\top\mathbf{z} = 0$, $\mathbf{z}^\top\omega = \mathbf{z}^\top\mathbf{Bf} = (\mathbf{B}^\top\mathbf{z})^\top\mathbf{f} = 0$. Conversely, if $\mathbf{z}^\top\omega = 0$ for all circulations, the maximum dual value is 0, which by Theorem LpC-T3 implies $\Phi_p([\omega]) = 0$. By Theorem LpC-T2, $[\omega] = 0$, so $\omega \in \text{im } d_0$.

□

Corollary L1C-C1 — exactness criterion (p = 1 base case)

ω is exact if and only if every circulation pairs to zero with ω.

Proof. Specialize Corollary LpC-C1 to $p = 1$.

□

4. Geometric transition across regimes

The geometry of the quotient problem transitions across the parameter $p \in [1, \infty]$. We classify these into distinct regimes:

4.1 Polyhedral regime (p = 1)

At $p = 1$, the objective is piecewise linear and non-strictly convex. Primal minimizers may be non-unique, forming a convex polyhedral set. Optimal pairs satisfy checkable certificate properties:

Theorem L1C-T4 — primal-dual certificate

Let $\mathbf{r} = \omega + \mathbf{Bf}$. Suppose $\mathbf{B}^\top\mathbf{z} = 0$, $|z_e| \leq w_e$, and:

- $z_e = w_e \operatorname{sgn}(r_e)$ whenever $r_e \neq 0$;
- $z_e \in [-w_e, w_e]$ when $r_e = 0$.

Then $\mathbf{r}$ is primal optimal, $\mathbf{z}$ is dual optimal, and

$$\Phi_1([\omega]) = \mathbf{z}^\top\omega = \|\mathbf{r}\|_{1,\mathbf{w}}.$$

Proof. Feasibility gives $\mathbf{z}^\top\omega = \mathbf{z}^\top(\omega + \mathbf{Bf}) = \mathbf{z}^\top\mathbf{r}$. The coordinate conditions give $\mathbf{z}^\top\mathbf{r} = \sum_{e\in E} w_e|r_e|$. Weak duality sandwiches this common value between dual and primal optima, establishing optimality of both. Lean 4 formalizes cycle3 complementary slackness in `lean4/l1c_t04_complementary_slackness.lean`.

□

Theorem L1C-TG — general finite matrix certificate

Let $m \in \mathbb{N}$ be an arbitrary finite dimension, $\mathbf{w} \in \mathbb{N}_{>0}^m$ positive integer weights, $\omega \in \mathbb{Z}^m$ an input 1-cochain, and $\mathbf{D} : \mathbb{Z}^k \to \mathbb{Z}^m$ an arbitrary discrete coboundary operator. If a candidate potential $\mathbf{p} \in \mathbb{Z}^k$, candidate dual vector $\mathbf{z} \in \mathbb{Z}^m$, and candidate primal residual $\mathbf{r} \in \mathbb{Z}^m$ satisfy:

1. Primal feasibility: $\mathbf{r} = \omega + \mathbf{Dp}$;
2. Exact gauge annihilation: $\langle \mathbf{z}, \mathbf{Dp}' \rangle = 0$ for all potentials $\mathbf{p}' \in \mathbb{Z}^k$;
3. Capacity bounds: $|z_e| \le w_e$ for all $e \in \{1, \dots, m\}$;
4. Zero duality gap: $\|\mathbf{r}\|_{1,\mathbf{w}} = \langle \mathbf{z}, \omega \rangle$; Then $\mathbf{r}$ is globally optimal across all affine gauge choices:

$$\|\mathbf{r}\|_{1,\mathbf{w}} \le \|\omega + \mathbf{Dp}'\|_{1,\mathbf{w}} \quad \forall \mathbf{p}' \in \mathbb{Z}^k.$$

Proof. For any $\mathbf{p}' \in \mathbb{Z}^k$, gauge annihilation yields $\langle \mathbf{z}, \omega + \mathbf{Dp}' \rangle = \langle \mathbf{z}, \omega \rangle + \langle \mathbf{z}, \mathbf{Dp}' \rangle = \langle \mathbf{z}, \omega \rangle$. The coordinate capacity bound $|z_e| \le w_e$ ensures $\langle \mathbf{z}, \mathbf{y} \rangle \le \|\mathbf{y}\|_{1,\mathbf{w}}$ for all $\mathbf{y} \in \mathbb{Z}^m$. Setting $\mathbf{y} = \omega + \mathbf{Dp}'$, we obtain:

$$\|\mathbf{r}\|_{1,\mathbf{w}} = \langle \mathbf{z}, \omega \rangle = \langle \mathbf{z}, \omega + \mathbf{Dp}' \rangle \le \|\omega + \mathbf{Dp}'\|_{1,\mathbf{w}}.$$

Mechanized in Lean 4 (`lean4/l1c_general_certificate.lean`) for arbitrary dimension m with zero `sorry` and kernel axioms `{propext, Classical.choice, Quot.sound}`.

□

4.2 Strictly and uniformly convex regime (p in (1, inf))

For $p \in (1, \infty)$, the L^p norm is strictly convex and uniformly convex by Clarkson's inequalities.

Theorem LpC-T4P — uniform convexity and nonlinear graph p-Laplacian

For any $p \in (1, \infty)$ and $\omega \in C^1$:

1. The primal minimizer is strictly unique on the gauge-fixed subspace $F = \{f \in C^0 : f(v_j) = 0 \text{ for one root } v_j \text{ per component}\}$, yielding a strictly unique optimal residual $\mathbf{r}_p^* = \omega + \mathbf{Bf}_p^*$.
2. The optimal potential $\mathbf{f}_p^*$ is characterized by the nonlinear graph p-Laplacian equation:

$$\mathbf{B}^\top \left(\mathbf{w} \odot |\mathbf{Bf} + \omega|^{p-1} \odot \operatorname{sgn}(\mathbf{Bf} + \omega) \right) = \mathbf{0}.$$

3. When the optimal residual is nonzero, its normalized dual witness is uniquely determined by:

$$\mathbf{z}_p^* = \frac{\mathbf{w} \odot |\mathbf{r}_p^*|^{p-1} \odot \operatorname{sgn}(\mathbf{r}_p^*)}{\|\mathbf{r}_p^*\|_{p,\mathbf{w}}^{p-1}}.$$

Proof. For (1), uniform convexity of the L^p norm on $\mathbb{R}^m$ for $p \in (1, \infty)$ establishes that for any $\mathbf{x} \neq \mathbf{y}$ with $\|\mathbf{x}\|_{p,\mathbf{w}} = \|\mathbf{y}\|_{p,\mathbf{w}} = 1$, $\|\frac{\mathbf{x}+\mathbf{y}}{2}\|_{p,\mathbf{w}} < 1$. The powered objective is strictly convex on the gauge-fixed space: the coboundary is injective there and each positive-weight scalar function $|x|^p$ is strictly convex. Attainment therefore gives uniqueness. For (2), for $p \in (1, \infty)$, the function $J_p(\mathbf{f}) = \frac{1}{p}\|\omega + \mathbf{Bf}\|_{p,\mathbf{w}}^p = \frac{1}{p}\sum_{e\in E} w_e|(\omega + \mathbf{Bf})_e|^p$ is strictly convex and continuously differentiable. Its gradient with respect to $\mathbf{f}$ is:

$$\nabla J_p(\mathbf{f}) = \mathbf{B}^\top \left(\mathbf{w} \odot |\omega + \mathbf{Bf}|^{p-1} \odot \operatorname{sgn}(\omega + \mathbf{Bf}) \right).$$

Setting $\nabla J_p(\mathbf{f}) = \mathbf{0}$ yields the nonlinear graph p-Laplacian Euler-Lagrange equation. For (3), setting $\mathbf{z}_p^*$ as displayed, we observe $\mathbf{B}^\top \mathbf{z}_p^* = \mathbf{0}$ (by the Euler-Lagrange equation), and direct evaluation confirms $\|\mathbf{z}_p^*\|_{q,\mathbf{w}^*} = 1$ and $\langle \mathbf{z}_p^*, \omega \rangle = \|\mathbf{r}_p^*\|_{p,\mathbf{w}}$, satisfying the Fenchel-Rockafellar optimality conditions with zero duality gap.

□

4.3 Classical Hilbertian regime (p = 2)

At $p = 2$, the L^2 geometry coincides with the classical Hilbertian framework.

Theorem LpC-T4H — Hilbertian L2 Hodge-Helmholtz decomposition

For $p = 2$ and uniform weights $w_e = 1$:

1. The cochain space admits an orthogonal Hodge-Helmholtz decomposition:
$$C^1 = \text{im } d_0 \oplus \ker(d_0^*) = B^1 \oplus Z_1.$$
2. The primal minimizer is given explicitly by the Moore-Penrose pseudoinverse of the standard graph Laplacian $\mathbf{L}_0 = \mathbf{B}^\top\mathbf{B}$:
$$\mathbf{f}_2^* = -\mathbf{L}_0^\dagger\mathbf{B}^\top\omega.$$
3. The optimal residual is the harmonic component $\mathbf{r}_2^* = (\mathbf{I} - \mathbf{B}\mathbf{L}_0^\dagger\mathbf{B}^\top)\omega$. When $\mathbf{r}_2^* \neq \mathbf{0}$, the dual circulation witness is self-dual:
$$\mathbf{z}_2^* = \frac{\mathbf{r}_2^*}{\|\mathbf{r}_2^*\|_2};$$
when $\mathbf{r}_2^* = \mathbf{0}$, the quotient norm vanishes $\Phi_2([\omega]) = 0$ and $\mathbf{z}_2^* = \mathbf{0}$ trivially satisfies dual feasibility and zero duality gap.

Proof. At $p = 2$, the p-Laplacian becomes linear: $\mathbf{B}^\top(\mathbf{B}\mathbf{f} + \omega) = \mathbf{L}_0\mathbf{f} + \mathbf{B}^\top\omega = \mathbf{0}$. Because $\mathbf{B}^\top\omega$ lies in the range of $\mathbf{L}_0$, the minimum-norm potential is $\mathbf{f}_2^* = -\mathbf{L}_0^\dagger\mathbf{B}^\top\omega$. Orthogonality of im d_0 and $\ker(d_0^*)$ gives Pythagoras' theorem: $\|\omega\|_2^2 = \|\mathbf{B}\mathbf{f}_2^*\|_2^2 + \|\mathbf{r}_2^*\|_2^2$. The dual exponent is $q = 2$, so dual circulations are measured in the same L^2 norm, making the dual witness $\mathbf{z}_2^*$ a direct scalar multiple of the primal harmonic residual when $\mathbf{r}_2^* \neq \mathbf{0}$. If $\mathbf{r}_2^* = \mathbf{0}$, the cochain is exact, $\Phi_2([\omega]) = 0$, and $\mathbf{z}_2^* = \mathbf{0}$ satisfies $\mathbf{B}^\top\mathbf{0} = \mathbf{0}$ with $\langle\mathbf{0}, \omega\rangle = 0$.

□

Positive nonuniform weights. Put $\mathbf{W} = \text{diag}(w_e)$ and $\mathbf{L}_\mathbf{w} = \mathbf{B}^\top\mathbf{W}\mathbf{B}$. The minimum Euclidean-norm potential is $-\mathbf{L}_\mathbf{w}^\dagger\mathbf{B}^\top\mathbf{W}\omega$, and stationarity is $\mathbf{B}^\top\mathbf{W}r_2^* = 0$. At nonzero residual the dual witness is

$$z_2^* = \frac{\mathbf{W}r_2^*}{\|r_2^*\|_{2,\mathbf{w}}},$$

with $\sum_e (z_e^*)^2/w_e = 1$ and $\langle z_2^*, \omega\rangle = \|r_2^*\|_{2,\mathbf{w}}$. At zero residual use $z_2^* = 0$. On the unit-flux directed triangle with weights $(1, 2, 3)$, $r_2^* = (6, 3, 2)/11$ is not a real circulation, whereas $\mathbf{W}r_2^* = (6, 6, 6)/11$ is. The weight factor is essential.

Contrast between L1 and L2 localization. While L^2 minimization distributes error across the entire graph to minimize squared Euclidean length (variance), L^1 minimization penalizes magnitude linearly, concentrating error onto sparse topological cycles. Section 8 provides an explicit instance where L^2 spreads defect across all edges while L^1 confines it to a single edge.

4.4 Chebyshev regime (p = inf)

At $p = \infty$, the quotient defect measures the minimal maximum-edge violation:

$$\Phi_\infty([\omega]) = \inf_f \max_{e\in E} w_e|(d_0 f + \omega)_e|.$$

Theorem LpC-T4INF — discrete Chebyshev approximation and active-edge pairing

1. $\Phi_\infty([\omega])$ is attained and equals the Chebyshev approximation radius of ω on the subspace im d_0.
2. An optimum admits an active-edge subgradient $\mathbf{z} \in \mathbb{R}^E$ satisfying $\mathbf{B}^\top\mathbf{z} = \mathbf{0}$, $\sum_{e\in E} |z_e|/w_e \leq 1$, and $\mathbf{z}^\top(\omega + \mathbf{B}\mathbf{f}_\infty^*) = \|\omega + \mathbf{B}\mathbf{f}_\infty^*\|_{\infty,\mathbf{w}}$.

Proof. Writing the weighted maximum norm in epigraph form reduces the problem to a finite linear program; the stated active-edge subgradient conditions are its LP dual and complementary slackness optimality conditions (Theorem LpC-T3 at $p = \infty, q = 1$). The earlier assertion of a discrete Aronsson equation for arbitrary offset cochains does not follow from this argument and remains omitted.

□

4.5 The $p \to 1^+$ limit and secondary entropy minimization

Because the polyhedral L^1 quotient problem admits a non-strictly convex objective, the set of optimal primal potentials

$$\mathcal{M}_1 = \arg\min_{f \in F} \|\omega + d_0 f\|_{1,\mathbf{w}}$$

is in general a convex polytope rather than a singleton. In contrast, for every $p \in (1, \infty)$, strict convexity ensures that the L^p minimizer $f_p^* \in F$ is strictly unique. This establishes a continuous primal trajectory $p \mapsto f_p^*$ for $p > 1$.

Proposition LpC-P1 — secondary selection in the $p \to 1^+$ limit

Let $G = (V, E)$ be a connected finite graph with strictly positive weights $\mathbf{w}$, and let $\mathcal{M}_1 \subset F$ denote the non-empty compact convex set of gauge-fixed L^1 minimizers. For each $p \in (1, 2]$, let $f_p^* \in F$ be the unique minimizer of $J_p(f) = \frac{1}{p}\|\omega + d_0 f\|_{p,\mathbf{w}}^p$. Then:

1. Every accumulation point $f_1^* = \lim_{k\to\infty} f_{p_k}^*$ as $p_k \to 1^+$ belongs to $\mathcal{M}_1$.
2. On the relative interior of an optimal face where the active sign pattern $\operatorname{sgn}((\omega + d_0 f)_e)$ is invariant, the first-order asymptotic expansion of the objective as $p \to 1^+$ is:

$$\frac{1}{p}\sum_{e\in E} w_e |r_f(e)|^p = \|\omega + d_0 f\|_{1,\mathbf{w}} + (p-1)\left[E(f) - \|\omega + d_0 f\|_{1,\mathbf{w}}\right] + O((p-1)^2).$$

3. Accordingly, the trajectory f_p^* selects among L^1 minimizers by minimizing the weighted entropy functional

$$E(f) = \sum_{e\in E} w_e |r_f(e)| \log |r_f(e)|$$

(with the standard continuous convention $0 \log 0 = 0$) over $\mathcal{M}_1$. The functional $E(f)$ is strictly convex on $\mathcal{M}_1$, ensuring that the entropy-selected minimizer $f_1^* = \lim_{p\to 1^+} f_p^*$ is unconditionally unique and that the entire trajectory converges.

Proof. Let $W = \sum_e w_e$ and $M_0 = \max_e |\omega_e|$ (the empty-edge case is trivial). Weighted Hölder and optimality give, uniformly for $1 < p \le 2$,

$$\|r_p^*\|_{1,\mathbf{w}} \le W^{1-1/p}\|r_p^*\|_{p,\mathbf{w}} \le W^{1-1/p}\|\omega\|_{p,\mathbf{w}} \le W M_0.$$

Thus $\|\mathbf{B} f_p^*\|_{1,\mathbf{w}} \le W M_0 + \|\omega\|_{1,\mathbf{w}}$. The fixed inverse bound for $\mathbf{B}|_F$ controls $\|f_p^*\|$ independently of p. Together with compactness of $\mathcal{M}_1$, this places the trajectory and the optimal face in a common compact set $K \subset F$. On K, the residuals $r_f = \omega + d_0 f$ are uniformly bounded: $|r_f(e)| \le M < \infty$ for all $e \in E, f \in K$. For any fixed scalar $x \in [-M, M]$, the Taylor expansion of $t \mapsto \frac{1}{p}|x|^p$ around $p = 1$ gives:

$$\frac{1}{p}|x|^p = |x| + (p-1)\left[|x| \log |x| - |x|\right] + R(p, x),$$

under the standard continuous convention $0 \log 0 = 0$, where the remainder satisfies $|R(p, x)| \le C(p-1)^2$ uniformly for all $x \in [-M, M]$ for a uniform constant $C < \infty$. Summing over edges with weights $w_e > 0$, the expansion is uniform on K:

$$J_p(f) = \frac{1}{p}\|\omega + d_0 f\|_{p,\mathbf{w}}^p = \|\omega + d_0 f\|_{1,\mathbf{w}} + (p-1)\left[E(f) - \|\omega + d_0 f\|_{1,\mathbf{w}}\right] + R_p(f),$$

where $\sup_{f\in K}|R_p(f)| \le C'(p-1)^2$ with $C' = C\sum_e w_e < \infty$. Uniform convergence of $\|\cdot\|_{p,\mathbf{w}} \to \|\cdot\|_{1,\mathbf{w}}$ as $p \to 1^+$ implies Γ-convergence of the objectives, which ensures that all cluster points of f_p^* minimize the L^1 quotient norm, establishing (1). This uniform expansion on K establishes (2).

For (3), let $g \in \mathcal{M}_1 \subset K$ be an arbitrary gauge-fixed L^1 minimizer. By optimality of f_p^* for J_p on F, we have:

$$J_p(f_p^*) \le J_p(g).$$

Substituting the uniform expansion for both f_p^* and g:

$$\|\omega+d_0 f_p^*\|_{1,\mathbf{w}}+(p-1)\left[E(f_p^*)-\|\omega+d_0 f_p^*\|_{1,\mathbf{w}}\right]+R_p(f_p^*) \le \|\omega+d_0 g\|_{1,\mathbf{w}}+(p-1)\left[E(g)-\|\omega+d_0 g\|_{1,\mathbf{w}}\right]+R_p(g).$$

Because $g \in \mathcal{M}_1$, $\|\omega+d_0 g\|_{1,\mathbf{w}} = \Phi_1([\omega])$. Because $\Phi_1([\omega])$ is the minimum of $\|\cdot\|_{1,\mathbf{w}}$, we have $\|\omega+d_0 f_p^*\|_{1,\mathbf{w}} \ge \Phi_1([\omega]) = \|\omega+d_0 g\|_{1,\mathbf{w}}$, so the zeroth-order term $\|\omega+d_0 f_p^*\|_{1,\mathbf{w}}-\|\omega+d_0 g\|_{1,\mathbf{w}} \ge 0$ is non-negative. Subtracting this non-negative quantity preserves the inequality:

$$(p-1)\left[E(f_p^*)-\|\omega+d_0 f_p^*\|_{1,\mathbf{w}}\right]+R_p(f_p^*) \le (p-1)\left[E(g)-\Phi_1([\omega])\right]+R_p(g).$$

Dividing both sides by $p-1>0$:

$$E(f_p^*)-\|\omega+d_0 f_p^*\|_{1,\mathbf{w}}+\frac{R_p(f_p^*)}{p-1} \le E(g)-\Phi_1([\omega])+\frac{R_p(g)}{p-1}.$$

Because $|R_p(f)| \le C'(p-1)^2$ on K, the remainder quotients satisfy $\lim_{p\to1^+}\frac{R_p(f)}{p-1}=0$. Now let $p_k \to 1^+$ be any sequence such that $f_{p_k}^* \to f_1^* \in K$. By (1), $f_1^* \in \mathcal{M}_1$, so $\|\omega+d_0 f_1^*\|_{1,\mathbf{w}} = \Phi_1([\omega])$. Taking the limit $k\to\infty$ and using continuity of $E(f)$ and $\|\cdot\|_{1,\mathbf{w}}$ on K:

$$E(f_1^*)-\Phi_1([\omega]) \le E(g)-\Phi_1([\omega]).$$

Adding $\Phi_1([\omega])$ to both sides yields:

$$E(f_1^*) \le E(g) \quad \text{for all } g \in \mathcal{M}_1.$$

Thus, every cluster point f_1^* minimizes $E(f)$ over the compact convex set $\mathcal{M}_1$.

Next, we establish unconditional strict convexity of $E(f)$ on $\mathcal{M}_1$. No two L^1-optimal residuals $r_1, r_2 \in \mathcal{R}_1 = \{\omega + d_0 f : f \in \mathcal{M}_1\}$ can have opposite nonzero signs in any coordinate: if $(r_1)_e > 0$ and $(r_2)_e < 0$, then $|\frac{1}{2}(r_1)_e + \frac{1}{2}(r_2)_e| < \frac{1}{2}|(r_1)_e| + \frac{1}{2}|(r_2)_e|$. With strictly positive weights $w_e > 0$, the midpoint $\frac{1}{2}(r_1+r_2)$ would satisfy $\|\frac{1}{2}(r_1+r_2)\|_{1,\mathbf{w}} < \Phi_1([\omega])$, contradicting the minimality of $\Phi_1([\omega])$. Therefore, the entire optimal residual set $\mathcal{R}_1$ lies within a single closed coordinate orthant. On this closed orthant, the scalar function $\eta(t) = t\log t$ (with $0\log 0 = 0$) is strictly convex on $[0,\infty)$ since $\eta''(t) = 1/t > 0$ for $t > 0$. Because the affine map $f \mapsto \omega + d_0 f$ is injective on the gauge-fixed subspace F, $E(f)$ is strictly convex on the compact convex set $\mathcal{M}_1$. A strictly convex function on a non-empty compact convex set attains its minimum at a unique point $f_{\text{ent}}^* \in \mathcal{M}_1$. Because every cluster point f_1^* satisfies $E(f_1^*) \le E(g)$ for all $g \in \mathcal{M}_1$, all cluster points coincide with f_{ent}^*. Because the trajectory $\{f_p^*\}_{p\in(1,2]}$ is contained in the compact set K and has only one possible cluster point f_{ent}^*, the entire trajectory converges: $\lim_{p\to1^+} f_p^* = f_{\text{ent}}^*$.

□

5. Sparse optimal representatives

Theorem L1C-T5 — cycle-rank support bound

Every quotient class has an optimal representative $\mathbf{r}$ with $|\text{supp}(\mathbf{r})| \le \beta$.

Proof. Choose a full-row-rank matrix $\mathbf{A}$ whose row space is $\ker(\mathbf{B}^\top)$. Two cochains represent the same class exactly when $\mathbf{Ar} = \mathbf{A}\omega$. The problem is therefore basis pursuit:

$$\min_{\mathbf{r}^+,\mathbf{r}^-} \mathbf{w}^\top(\mathbf{r}^+ + \mathbf{r}^-) \quad \text{subject to } \mathbf{A}(\mathbf{r}^+ - \mathbf{r}^-) = \mathbf{A}\omega, \quad \mathbf{r}^+, \mathbf{r}^- \geq 0.$$

It has an optimal basic feasible solution. Since $\operatorname{rank}(\mathbf{A}) = \beta$, at most β variables among $\mathbf{r}^+$ and $\mathbf{r}^-$ are basic and positive. Positive weights exclude simultaneously positive r_e^+, r_e^-, since decreasing both preserves feasibility and lowers cost. Hence the resulting $\mathbf{r}$ has at most β nonzero coordinates.

□

This is an existential bound, not a genericity statement and not uniqueness. Degenerate instances may have dense optima as well as a sparse one.

Example — one directed cycle

For a consistently oriented m-cycle, the invariant is $S = \sum_{e\in E} \omega_e$. Hence

$$\Phi_1([\omega]) = |S| \min_{e\in E} w_e.$$

An optimum concentrates S on any minimum-weight edge. In the unweighted case the value is $|S|$, and the support bound $\beta = 1$ is sharp.

6. Integral cut-lattice optimum

Theorem L1C-T6 — integral attainment

If ω is integral and the weights are integral, there exists an optimal pair with integral $\mathbf{f}$ (after fixing one vertex gauge) and integral residual $\mathbf{r} = \omega + \mathbf{Bf}$.

Proof. Delete the root column of the incidence matrix. The resulting matrix is totally unimodular [2, 3]. Adjoining signed identity columns and column negations preserves total unimodularity, so the standard-form constraint matrix for the primal LP is totally unimodular. With integral right-hand side, an optimal basic feasible solution may be chosen integral.

□

This is a cochain/cut-lattice statement. It is not the minimum homologous-chain problem, whose repair directions are boundaries from the next chain degree.

7. Discrete physics and simplicial Bianchi nilpotency

On a finite 2-dimensional simplicial complex $K = (V, E, F)$, edge cochains model discrete gauge connections, while face cochains model discrete field curvature. Let $D_1 : C_1(K) \to C_0(K)$ and $D_2 : C_2(K) \to C_1(K)$ denote the standard signed simplicial boundary operators with respect to chosen orientations, and let $d_0 = D_1^\top : C^0(K) \to C^1(K)$ and $d_1 = D_2^\top : C^1(K) \to C^2(K)$ denote their algebraic coboundaries.

Theorem L1C-T7 — simplicial Bianchi nilpotency

Let $K = (V, E, F)$ be a finite 2-simplicial complex.

1. For every 0-cochain $f \in C^0(K)$, the composite coboundary vanishes identically:

$$d_1(d_0 f) = 0.$$

Consequently, pure gauge cochains have zero field curvature and lie in the cocycle subspace $Z^1(K) = \ker(d_1)$.

2. For every 1-cycle $z \in C_1(K)$ satisfying $D_1 z = 0$ (Kirchhoff circulation) and every 0-cochain $f \in C^0(K)$,

$$\langle z, d_0 f \rangle = 0.$$

The quotient norm $\Phi_1([\omega])$ is invariant under affine shifts along the gauge orbit: $\Phi_1([\omega + d_0 f]) = \Phi_1([\omega])$.

Proof. For (1), by definition of the adjoint, $(d_1 \circ d_0)^\top = D_1 \circ D_2$. For each oriented triangular face $\sigma = (u, v, w) \in F$, the boundary is $D_2\sigma = (u, v) + (v, w) - (u, w)$. Applying D_1 yields

$$D_1(D_2\sigma) = (v - u) + (w - v) - (w - u) = 0.$$

Because $D_1 D_2 = 0$, its transpose $d_1 d_0 = (D_1 D_2)^\top = 0$.

For (2), the definition of the coboundary adjoint gives $\langle z, d_0 f \rangle = \langle D_1 z, f \rangle$. Because $D_1 z = 0$, this pairing evaluates to zero. Invariance of Φ_1 follows because $\omega + d_0 f + d_0 g = \omega + d_0(f + g)$, which parametrizes the identical coset $[\omega] \in Q^1$.

□

8. Quantum phase cocycles and Schatten-p operator extensions

Let $G = (V, E)$ be a finite directed graph whose vertices index an orthonormal basis $\{|v\rangle\}_{v \in V}$ of an n-dimensional Hilbert space $\mathcal{H} = \mathbb{C}^V$. Let $\omega \in C^1(G, \mathbb{R})$ denote a phase 1-cochain assigning phase difference $\omega(u, v)$ to each directed edge $(u, v) \in E$. A quantum state on $\mathcal{H}$ is described by a density matrix $\rho = \sum_{u,v \in V} \rho_{uv} |u\rangle\langle v|$ satisfying $\rho = \rho^\dagger$, $\mathrm{Tr}(\rho) = 1$, and $\rho \succeq 0$. Let $\rho_{\mathrm{diag}} = \sum_{v \in V} \rho_{vv} |v\rangle\langle v|$ denote the dephased pointer state.

For operators on $\mathcal{H}$, the Schatten-p norm for $p \in [1, \infty)$ is $\|A\|_p = (\mathrm{Tr}(|A|^p))^{1/p}$, and for $p = \infty$, $\|A\|_\infty = \|A\|_{\mathrm{op}}$. For $p = 1$, this specializes to the trace norm $\|A\|_1 = \mathrm{Tr}\,|A| = \mathrm{Tr}\,\sqrt{A^\dagger A}$.

Theorem LpC-T8 — Schatten-p quantum defect and non-commutative Hölder bound

1. For any operator-valued cochain $H \in C^1(G, \mathcal{B}_p(\mathcal{H}))$, the Schatten-$p$ quotient defect is:

$$\Phi_p([H]) = \inf_{F \in C^0(G, \mathcal{B}_p(\mathcal{H}))} \|H - d_0 F\|_{p, \mathbf{w}}.$$

It satisfies non-commutative Fenchel-Rockafellar Hölder duality:

$$\Phi_p([H]) = \max_{Z \in C_1(G, \mathcal{B}_q(\mathcal{H}))} \left\{ \mathrm{Re}\,\mathrm{Tr}\langle Z, H \rangle : d_0^* Z = 0, \quad \|Z\|_{q, \mathbf{w}^*} \leq 1 \right\},$$

where $1/p + 1/q = 1$ and non-commutative Hölder duality $|\,\mathrm{Tr}(A^\dagger B)| \leq \|A\|_p \|B\|_q$ holds.

2. Let $A = \rho - \rho_{\mathrm{diag}}$ be a $d \times d$ off-diagonal matrix. For $1 \leq p \leq 2$,

$$\|A\|_{S_p} \leq \left(\sum_{u \neq v} |A_{uv}|^p \right)^{1/p};$$

for $2 \leq p \leq \infty$,

$$\|A\|_{S_p} \leq d^{1-2/p} \left(\sum_{u \neq v} |A_{uv}|^p \right)^{1/p}.$$

The factor is interpreted as d at $p = \infty$. At $p = 1$ this recovers $\|\rho - \rho_{\mathrm{diag}}\|_1 \leq C_{l_1}(\rho)$.

Proof. For (1), non-commutative L^p spaces over finite-dimensional matrix algebras satisfy standard Clarkson inequalities and uniform convexity for $p \in (1, \infty)$. The dual space of the Schatten-p class $\mathcal{B}_p(\mathcal{H})$ is isometrically isomorphic to $\mathcal{B}_q(\mathcal{H})$ under the trace pairing $\langle A, B \rangle = \mathrm{Tr}(A^\dagger B)$ (see Watrous [6]). For (2), the $1 \leq p \leq 2$ bound follows by interpolation between the entrywise $\ell^1 \to S_1$ triangle inequality and the Frobenius identity. For $p \geq 2$, use $\|A\|_{S_p} \leq \|A\|_{S_2} = \|A\|_{\ell^2}$ and $\|A\|_{\ell^2} \leq d^{1-2/p} \|A\|_{\ell^p}$.

□

Theorem L1C-T8 — quantum phase cocycle defect bound (p = 1 base case)

All phase lifts, gradients, cycles and phase pairings in this theorem use the support graph $G_\rho = (V, E_\rho)$, where $E_\rho = \{(u,v) \in E : \rho_{uv} \neq 0\}$. Norms of the full density matrix and its coherence sum retain all matrix entries.

1. Let ρ be a density matrix with representation in the orthonormal basis $\{|v\rangle\}_{v\in V}$. Under the incidence convention $(d_0\theta)(u,v) = \theta_v - \theta_u$, the local diagonal unitary operator $U_\theta = \sum_{v\in V} e^{-i\theta_v}|v\rangle\langle v| = \mathrm{diag}(e^{-i\theta_v})$ acts on matrix entries as:

$$(U_\theta \rho U_\theta^\dagger)_{uv} = \langle u|U_\theta \rho U_\theta^\dagger|v\rangle = e^{-i\theta_u}\rho_{uv}e^{i\theta_v} = e^{i(\theta_v-\theta_u)}\rho_{uv} = e^{i(d_0\theta)_{uv}}\rho_{uv}.$$

   Thus, U_θ shifts each edge phase factor by $+(d_0\theta)_{uv}$. If ρ admits a chosen real phase lift ω satisfying $d_0\theta = -\omega$ (that is, $[\omega] = \mathbf{0} \in Q^1$), the gauge rotation U_θ eliminates all relative phase angles, yielding $(U_\theta\rho U_\theta^\dagger)_{uv} = |\rho_{uv}|$ without changing the spectrum or off-diagonal moduli $|\rho_{uv}|$.
2. The trace distance $\|\rho - \rho_{\mathrm{diag}}\|_{\mathrm{tr}} = \frac{1}{2}\,\mathrm{Tr}\,|\rho - \rho_{\mathrm{diag}}|$ is bounded by the total off-diagonal L^1 coherence:

$$\|\rho - \rho_{\mathrm{diag}}\|_{\mathrm{tr}} \leq \sum_{u<v} |\rho_{uv}| = \frac{1}{2}C_{l_1}(\rho),$$

   where $C_{l_1}(\rho) = \sum_{u\neq v}|\rho_{uv}|$ is the standard l_1-norm of coherence [5, 6]. For every edge $(u,v) \in E$, the 2×2 principal minor condition of positive semidefiniteness enforces

$$|\rho_{uv}| \leq \sqrt{\rho_{uu}\rho_{vv}} \leq \frac{1}{2}(\rho_{uu} + \rho_{vv}).$$

   For any nonzero real circulation $C \in Z_1(G;\mathbb{R})$, define its maximum weight-normalized capacity:

$$M_C = \max_{e\in E: C_e\neq 0} \frac{|C_e|}{w_e}.$$

   Then the normalized circulation $\mathbf{z} = \mathbf{C}/M_C$ is dual-feasible, establishing the lower bound:

$$\Phi_1([\omega]) \geq \frac{|\langle C,\omega\rangle|}{M_C} = \frac{|\langle C,\omega\rangle|}{\max_{e:C_e\neq 0}(|C_e|/w_e)}.$$

3. Explicit relationship between phase cochains and density operator ρ: Phase angle extraction is restricted to the support subgraph of non-vanishing coherences:

$$E_\rho = \{(u,v) \in E : \rho_{uv} \neq 0\}.$$

   For $(u,v) \in E \setminus E_\rho$, the phase angle is undefined. On E_ρ, any matrix entry factors as $\rho_{uv} = |\rho_{uv}|e^{i\omega(u,v)}$ for a chosen real lift $\omega \in C^1(E_\rho;\mathbb{R})$. Under diagonal unitary transformations $U_\theta = \mathrm{diag}(e^{-i\theta_v})$, entries transform as

$$(U_\theta\rho U_\theta^\dagger)_{uv} = |\rho_{uv}|e^{i(\omega(u,v)+(d_0\theta)_{uv})}.$$

   Crucially, taking principal arguments $\mathrm{Arg}(\cdot) \in (-\pi,\pi]$ edge-by-edge after a diagonal unitary rotation does not generally produce an exact gradient shift in $\mathbb{R}^E$, because branch wrapping introduces an integer period shift $\mathbf{k} \in \mathbb{Z}^E$:

$$\mathrm{Arg}((U_\theta\rho U_\theta^\dagger)_{uv}) = \omega(u,v) + (d_0\theta)_{uv} + 2\pi k_{uv}.$$

   On a triangle, start with positive real off-diagonal entries $\rho_{uv} > 0$, so initial edge phases and real quotient norm vanish ($\omega = \mathbf{0}, \Phi_1([\omega]) = 0$). Applying U_θ with $\theta = (0, 3\pi/4, 3\pi/2)$ gives potential differences $d_0\theta = (3\pi/4, 3\pi/4, -3\pi/2)$. Taking principal branch arguments wraps the third edge phase to $\pi/2 = -3\pi/2 + 2\pi$, producing principal phases $(3\pi/4, 3\pi/4, \pi/2)$. The sum around the directed cycle is $3\pi/4 + 3\pi/4 + \pi/2 = 2\pi$, so the re-extracted real quotient norm is $\Phi_1 = 2\pi$, even though the state underwent only a local diagonal unitary rotation and the physical $U(1)$ holonomy around the cycle remains $\exp(i\cdot 2\pi) = 1$. Therefore, three quantities must be distinguished:

(a) **Chosen real lift quotient norm:** For a fixed real lift $\omega \in C^1(E_\rho;\mathbb{R})$, $\Phi_1([\omega]) = \inf_{f\in C^0} \|\omega + d_0 f\|_{1,\mathbf{w}}$ is invariant under real gradient shifts $\omega \mapsto \omega + d_0 f$, measuring minimal L^1 effort to gauge-align that specific real lift.

(b) **Gauge-invariant $U(1)$ cycle holonomy:** On the support graph $G_\rho = (V, E_\rho)$, for any integral cycle $C \in Z_1(G_\rho;\mathbb{Z})$, the multiplicative holonomy $\mathrm{hol}_{U(1)}(C) = \prod_{e\in E_\rho} (\rho_{uv}/|\rho_{uv}|)^{C_e} = \exp(i\langle C, \omega\rangle) \in U(1)$ is strictly invariant under all diagonal unitary transformations $U_\theta \rho U_\theta^\dagger$, since $\exp(i\langle C, d_0\theta\rangle) = \exp(i\langle D_1 C, \theta\rangle) = 1$.

(c) **Branch-independent $U(1)$ quotient defect:** To obtain a quotient distance from the identity on $U(1)$ connections that is invariant under diagonal unitaries without branch-cut artifacts, one minimizes over both vertex potentials and integer edge branch shifts ($2\pi\mathbb{Z}^{E_\rho}$):

$$\Phi_1^{U(1)}(\rho) = \inf_{f\in C^0(G_\rho),\, \mathbf{k}\in\mathbb{Z}^{E_\rho}} \|\omega + d_0 f + 2\pi\mathbf{k}\|_{1,\mathbf{w}}.$$

This is a length function on the quotient group $\mathbb{R}^{E_\rho}/(\mathrm{im}\, d_0 + 2\pi\mathbb{Z}^{E_\rho})$, inducing the distance $d([\omega],[\eta]) = \Phi_1^{U(1)}([\omega - \eta])$. It is not a norm on a real vector space: on an unweighted cycle, flux π has length π, while doubled flux 2π has length zero. Integral coefficients in (b) ensure independence of phase lift; a real half-cycle changes holonomy by -1 after a 2π lift change on one edge. Real circulations remain the domain of the linear LP dual in (a). For the triangle counterexample above, choosing $\mathbf{k} = (0, 0, -1)$ and $f = -\theta$ cancels the branch winding, giving $\Phi_1^{U(1)}(\rho) = 0$.

Proof. Specializing Theorem LpC-T8 to $p = 1$, direct evaluation under diagonal unitary conjugation yields (1). For (2), the triangle inequality across 2×2 off-diagonal blocks gives the trace distance bound. The capacity-normalized loop bound follows from weak LP duality with dual-feasible vector $\mathbf{z} = \mathbf{C}/M_C$, tested in `test_theorem_8_weighted_loop_bound_correction`. For (3), restricting phase extraction to E_ρ avoids undefined arguments at zero coherences. The sign $+(d_0\theta)_{uv}$ follows from $(U_\theta \rho U_\theta^\dagger)_{uv} = e^{-i\theta_u}\rho_{uv}e^{i\theta_v} = e^{i(\theta_v - \theta_u)}\rho_{uv} = e^{i(d_0\theta)_{uv}}\rho_{uv}$. The branch-cut wrapping and the distinction between real lifts and $U(1)$ holonomy are validated on the triangle counterexample $\theta = (0, 3\pi/4, 3\pi/2)$ in `test_quantum_phase_gauge_invariance_and_branch_winding` and `test_quantum_phase_additive_flux_vs_u1_holonomy`. In this finite discrete setting, we emphasize that these combinatorial bounds on off-diagonal matrix entries are static geometric inequalities, separate from continuous unitary Hamiltonian time evolution.

□

9. Cohomological classification and topological obstructions

For a general cell complex, $H^1(K;\mathbb{R}) = \ker(d_1)/\,\mathrm{im}(d_0)$. The quotient norm Φ_p is defined on $C^1/\,\mathrm{im}(d_0)$ and descends to H^1 only after restricting to closed cochains. On a graph, $C^2 = 0$ and this distinction disappears:

1. **Exact 1-Cochains ($[\omega] = 0$):** $\Phi_p([\omega]) = 0$ if and only if $\omega \in B^1 = \mathrm{im}\, d_0$. The cochain is a pure gradient potential ($r = 0$), corresponding to a trivial gauge connection with zero holonomy along all cycles.
2. **Closed Cohomology Defect ($[\omega] \neq 0$ on 1-graphs):** On a 1-dimensional graph complex ($d_1 = 0$), all cochains are cocycles ($Z^1 = C^1$). When $\Phi_p([\omega]) > 0$, ω represents a non-trivial cohomology class in $H^1(G;\mathbb{R}) \cong \mathbb{R}^\beta$. The defect represents a topological obstruction that cannot be eliminated by any choice of vertex potential.
3. **Simplicial Curvature Obstructions (2-complexes):** On a 2-simplicial complex K, the face coboundary $d_1\omega \in C^2(K)$ measures discrete field curvature (magnetic flux). If $d_1\omega \neq 0$, ω is not closed ($\omega \notin Z^1$), indicating localized curvature defect on 2-simplices. By Theorem L1C-T7 (simplicial Bianchi nilpotency), any exact gauge adjustment $d_0 f$ has zero curvature ($d_1(d_0 f) = 0$), so the curvature $d_1\omega$ is strictly gauge-invariant.

This taxonomy classifies discrete difference fields by their topological obstructions, distinguishing exact gauge transformations, non-contractible cycle circulations, and non-vanishing face curvatures.

10. Computational companion & formal mechanization

The companion software implementation provides exact rational arithmetic for all polyhedral and certificate decision paths ($p = 1, 2, \infty$, $LDL^\dagger$ positive-semidefiniteness, and simplicial complexes) under the Zero-Float Law, avoiding numerical tolerances and floating-point ambiguities, while continuous interior-p iterations and Schatten-p singular values are evaluated via binary64 numerical diagnostics:

1. **Exact Two-Phase Simplex Solver**: An exact rational linear programming solver over `fractions.Fraction` implementing Bland's minimal-index pivot rule, guaranteeing cycle prevention for the $p = 1$ base case. It handles primal standard forms, artificial variable elimination, and dual variable extraction without floating-point error, and raises `SolverIterationLimitError` on budget exhaustion.
2. **Primal-Dual Certification**: Checks primal feasibility $r = \omega + d_0 f$, dual divergence-free annihilation $\mathbf{B}^\top \mathbf{z} = 0$, capacity bounds $|z_e| \le w_e$, and verifies exact zero duality gap $\Phi_1([\omega]) - \mathbf{z}^\top \omega = 0$. Exposes standalone receipt validator `check_certificate_receipt`; its integrality flag concerns the primal potential and residual, not the dual. The Chebyshev path independently checks $r = h - \mathbf{B}f$, $t \ge 0$, $w_e|r_e| \le t$, $\mathbf{B}^\top z = 0$, $\sum_e |z_e|/w_e \le 1$, and $\langle z, h\rangle = t$ through `check_linf_certificate_receipt` before returning `OPTIMAL_LINF`. The opposite residual sign is a solver convention and preserves the gauge quotient.
3. **Simplicial Complex & Bianchi Operators**: Implements boundary matrices D_1, D_2, coboundaries d_0, d_1, and automated verification of $d_1(d_0 f) = 0$, distinguishing closed graph residuals from curvature residuals.
4. **Quantum Density Matrix Engine**: Pure standard-library module verifying Hermiticity, trace normalization, and exact positive semidefiniteness via rational $LDL^\dagger$ decomposition with complete diagonal pivoting. Coherence values use exact fractions when all required complex moduli are rational, and binary64 square roots otherwise. The L^2 squared norm is exact; its displayed square root may be numerical.
5. **Interactive Notebooks**: Dual-track Jupyter notebooks (`001_conceptual_l1_cohomology.ipynb` and `002_formal_l1_cohomology.ipynb`) provide pedagogical visualizations, defect injection sandboxes, and tableau inspections.
6. **Formal Interactive Proof Mechanization**: Lean 4 mechanizes `general_finite_certificate_optimality` and the named finite-model theorems for L1C-T1, L1C-T2, L1C-T3, L1C-T4, L1C-T5, and L1C-T7. Integer 3-cycle lemmas in `l1c_lp_duality`, `l1c_lp_strict_uniqueness`, and `l1c_lp_hodge_decomposition` do not mechanize the general real L^p propositions. The Lean estate compiles offline with `lake build`, with proof-placeholder commands absent and axiom reports inside `{propext, Classical.choice, Quot.sound}`.

The shipped pytest suite contains 91 test functions (producing 110 test cases) across trees, cycles, complete graphs, disconnected graphs, zero-weight refusal, integer lattices, simplex iteration limits, primal feasibility fault injections, exact $LDL^\dagger$ PSD checks, quantum phase gauge invariance and branch-cut winding checks, and the L^1/L^2 counterexample $\mathbf{B} = (1, 1, 1)^\top$, $\omega = (0, 0, 10)^\top$. Replay of the complete test suite and Lean proof verification is invoked explicitly via: `python verify_deposit.py --run-tests --lean`. Current replay of those tests requires the optional development dependencies `pytest` and `pytest-cov`; source compilation and Lean replay are recorded separately from any historical coverage receipt.

11. Scope and concluding remarks

This monograph investigates finite-dimensional quotient spaces of discrete cell complexes under the family of weighted L^p norms. We summarize the precise boundaries of these results:

1. **Discrete vs. Continuum:** The results herein apply to finite directed graphs and finite 2-simplicial complexes. They do not constitute an infinite-dimensional Hodge-de Rham theorem on manifolds, nor do they address continuous Sobolev or L^p cohomology on non-compact domains.
2. **Static vs. Dynamical:** The bounds on quantum density matrix coherence (Theorems LpC-T8 and L1C-T8) apply to fixed states in a chosen basis; they do not characterize continuous unitary time-evolution or open quantum system master equations.
3. **Formal Verification:** Lean 4 mechanizes `general_finite_certificate_optimality` and the named finite-model theorems for L1C-T1, L1C-T2, L1C-T3, L1C-T4, L1C-T5, and L1C-T7. The general L^p results, exactness characterization, cycle-rank bound, total unimodularity, and Schatten statements have prose

proofs and are classified as classical background. They are not mechanized at that generality. L1C-T2 mechanizes ambient integer three-edge weighted-norm axioms and residual-addition compatibility, without defining a minimized quotient norm. The conditional finite integer certificate establishes optimality from its hypotheses; it does not establish certificate existence or verify the Python implementation. The earlier general Aronsson assertion for arbitrary cochains remains withdrawn.

References

1. R. T. Rockafellar, *Convex Analysis*, Princeton University Press, 1970.
2. A. Schrijver, *Theory of Linear and Integer Programming*, Wiley, 1986.
3. T. K. Dey, A. N. Hirani, and B. Krishnamoorthy, "Optimal homologous cycles, total unimodularity, and linear programming," *SIAM Journal on Computing* 40(4), 2011, 1026–1044. https://doi.org/10.1137/100800245
4. S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004. https://web.stanford.edu/~boyd/cvxbook/
5. T. Baumgratz, M. Cramer, and M. B. Plenio, "Quantifying Coherence," *Physical Review Letters* 113(14), 2014, 140401. https://doi.org/10.1103/PhysRevLett.113.140401
6. J. Watrous, *The Theory of Quantum Information*, Cambridge University Press, 2018. https://doi.org/10.1017/9781316848142
7. J. A. Clarkson, "Uniformly convex spaces," *Transactions of the American Mathematical Society* 40(3), 1936, 396–414. https://doi.org/10.1090/S0002-9947-1936-1501880-4
8. G. Aronsson, "Extension of functions satisfying Lipschitz conditions," *Arkiv för Matematik* 6(6), 1967, 551–561. https://doi.org/10.1007/BF02591361
9. X. Jiang, L.-H. Lim, Y. Yao, and Y. Ye, "Statistical ranking and combinatorial Hodge theory," *Mathematical Programming* 127(1), 2011, 203–244. DOI (preprint: arXiv:0811.1067v2, §6.1).